

David Marr

Lead Software Engineer — Data Platforms, Analytics & Delivery

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Summary

Engineering leader who builds reliable data platforms and services — turning loosely defined product and analytics requirements into well-modeled, scalable systems teams can trust for reporting and day-to-day operations.

Twenty-five years shipping production software on Python, Java, AWS, and PostgreSQL. I lead in Agile teams, mentor engineers, and raise the bar on data quality, observability, and predictable delivery.

Experience

Context Labs, Principal Software Engineer / Engineering Lead (Platform) – Cambridge, MA June 2022 – Mar 2026

Led distributed engineering organization delivering a multi-tenant analytics platform — owning how data is modeled, accessed, and delivered across product, reporting, and geospatial use cases.

- Managed and grew engineering team across three continents in Scrum delivery; introduced OKRs and operational reviews that improved velocity and cut incident recovery time by 50%.
- Architected production data-service layer (GraphQL/Java) with multi-tenancy, single sign-on, role-based access, and internationalization — the integration point for analytics, dashboards, and product features.
- Partnered with product and analytics stakeholders to shape relational data models and APIs supporting reporting, charting (Vega Lite), and map-based visualization (Mapbox).
- Established engineering standards — code review, testing, and monitoring practices — that improved data trust and reduced operational friction across the platform.
- Mentored engineers and drove adoption of AI-assisted development workflows with hands-on workshops and clear quality guardrails.

Skills used: Python, Java, GraphQL, PostgreSQL, Trino, TypeScript, Vue, Nuxt, Mapbox

Filtered.ai, Lead Engineer – Boston, MA

Feb 2020 – Feb 2022

Lead engineer building AWS-backed data and ML workflows for technical hiring assessments — ingestion, validation, scoring, and reporting for enterprise recruiting teams.

- Built fraud and integrity detection pipelines using Python, computer vision, and AWS (including Mechanical Turk for human-in-the-loop validation).
- Owned full-stack hiring platform on PostgreSQL — assessment data models, scheduling, grading workflows, and operational reporting.
- Collaborated with product and data science on model integration, quality checks, and reliable delivery of automated scoring at scale.

Skills used: Python, AWS, PostgreSQL, Node.js, TypeScript, React

Turo, Staff Software Engineer – Cambridge, MA

Apr 2018 – Dec 2019

Owned trip-management and dynamic pricing systems on PostgreSQL, improving host experience and fleet utilization through data-driven availability and rate logic.

- Maintained the All-star host program to recognize and incentivize top hosts, supporting retention and higher-quality listings.
- Partnered with product and operations on pricing and availability rules that depended on reliable transactional and reporting data.

Skills used: PostgreSQL, React, Redux, TypeScript, JavaScript

OneMarket / Westfield Labs, Senior Software Architect – San Francisco, CA Jan 2016 – May 2018
Integrated facial recognition, subscription billing, and automated parking gates into a smart parking platform with payment and usage data flowing through PostgreSQL-backed services.

- Architected physical and online payment products, yielding ~\$2.7m in monthly recurring revenue with reliable billing and reporting data.
- Built indoor mapping, directory kiosks, and wayfinding systems across 4 flagship shopping centers, averaging 60,000+ monthly sessions with operational metrics for venue stakeholders.

Skills used: PostgreSQL, Node.js, React, JavaScript, Stripe, Mapbox, C

Tilt.com, Senior Systems Engineer – San Francisco, CA July 2013 – Dec 2015

- Implemented and iterated a simplified contribution flow yielding 15% lift in site signups.
- Pioneered server rendered React using reification with Perl/Dancer backend.

Skills used: React, Perl, JavaScript, PostgreSQL

Various, Previous Leadership & Technical Roles – Massachusetts & California 2000-2013

- Publishing and ad-tech at Say Media (audience data, campaign delivery, and content platforms); co-founded early social product (Secret Feature).
- Design-to-production prototyping (Frog Design); scaled frontend architecture for high-traffic Yahoo properties.
- Webby-recognized career advice site (Monster.com); enterprise web integrations (SoftArtisans).
- Industrial automation web interfaces and dashboards (GE Fanuc / Intellution)

Skills used: Node.js, Adobe Flash/Flex/FMS, MySQL, PHP, SQL Server, .NET, ASP Classic, HTML, CSS, JavaScript

Education

University of New Hampshire, BS in Business Administration – Durham, NH Sept 1996 – May 2000
Focus on Management of Information Systems (MIS)

Skills

Data & Platform: Python, Java, PostgreSQL, AWS, Trino, GraphQL, Docker, Kubernetes, MongoDB, Bash, Git

Languages/Frameworks/Web: TypeScript, JavaScript, Go, React, Vue, HTML, CSS, Tailwind, Nuxt, Tanstack Start, Vega Lite, Mapbox, Maplibre, Deck.gl, D3, Zod, React Hook Form, Vee-Validate, WebGL, WebGPU, WebRTC

AI Tools/Infrastructure: Vite, Vitest, Vercel AI SDK, Pi, Cursor, Opencode, Cloudflare Workers, Cloudflare Durable Objects, Agent Skills, Turborepo, Lerna/NX, ArcadeDB